

$$1. \mathbb{Z}_{16} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}.$$

$$\mathbb{Z}_{16} = \langle 1 \rangle$$

满足 $(16, r) = 1$ 的 $r = 1, 3, 5, 7, 9, 11, 13, 15$

$\therefore \mathbb{Z}_{16}$ 的生成元有 $1, 3, 5, 7, 9, 11, 13, 15$.

16 的正因子有 $1, 2, 4, 8, 16$.

$\therefore \mathbb{Z}_{16}$ 的子群有 $\langle 1 \rangle, \langle 2 \rangle, \langle 4 \rangle, \langle 8 \rangle, \langle 16 \rangle$

$$\langle 1 \rangle = \mathbb{Z}_{16}$$

$$\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14\}$$

$$\langle 4 \rangle = \{0, 4, 8, 12\}$$

$$\langle 8 \rangle = \{0, 8\}$$

$$\langle 16 \rangle = \{0\}$$

$$\text{ord } 1 = \frac{16}{(16, 1)} = 16$$

$$\text{ord } 2 = \frac{16}{(16, 2)} = 8$$

$$\text{ord } 3 = \frac{16}{(16, 3)} = 16$$

$$\text{ord } 4 = \frac{16}{(16, 4)} = 4$$

$$\text{ord } 6 = \frac{16}{(16, 6)} = 8$$

$$\text{ord } 8 = \frac{16}{(16, 8)} = 2$$

$$\text{ord } 10 = \frac{16}{(16, 10)} = 8$$

$$\text{ord } 12 = \frac{16}{(16, 12)} = 4$$

$$\text{ord } 14 = \frac{16}{(16, 14)} = 8$$

$$\text{ord } 1 = \text{ord } 3 = \text{ord } 5 = \text{ord } 7 = \text{ord } 9 = \text{ord } 11 = \text{ord } 13 = \text{ord } 15 = 1.$$

$$2. U(18) = \{1, 5, 7, 11, 13, 17\} \quad |U(18)| = 6$$

$$\text{ord } 1 = 1$$

$$5^2 = 25 \equiv 7 \quad 5^3 = 35 \equiv 17 \quad 5^4 = 85 \equiv 13 \quad 5^5 = 65 \equiv 11 \quad 5^6 = 55 \equiv 1 \quad \therefore \text{ord } 5 = 6$$

$$7^2 = 49 \equiv 13 \quad 7^3 = 13 \times 7 = 91 \equiv 1 \quad \therefore \text{ord } 7 = 3$$

$$11^2 = 121 \equiv 13 \quad 11^3 = 11 \times 13 \equiv 17 \quad 11^4 = 17 \times 11 \equiv 7 \quad 11^5 = 77 \equiv 5 \quad 11^6 = 55 \equiv 1 \quad \text{ord } 11 = 6$$

$$13^2 = 169 \equiv 7 \quad 13^3 = 91 \equiv 1 \quad \text{ord } 13 = 3$$

$$17 = -1 \quad 17^2 \equiv 1 \quad \text{ord } 17 = 2$$

$\therefore \text{ord } 5 = \text{ord } 11 = |U(18)| = 6$ $\therefore$ 是循环群. 生成元是 $5, 11$

$$6 \text{ 的正因子 } d: 1, 2, 3, 6. \quad U(18) = \langle 5 \rangle$$

$\therefore G$ 的子群有 $\langle 5 \rangle, \langle 25 \rangle, \langle 125 \rangle, \langle 1 \rangle$ 即 $\langle 1 \rangle, \langle 5 \rangle, \langle 7 \rangle, \langle 17 \rangle$

即 $\{1\}, \{1, 5, 7, 11, 13, 17\}, \{1, 17\}, \{1, 7, 13\}.$



3. 由拉格朗日定理. G 是一个有限群, H 是 G 的子群.

存在性:

$$\text{则 } |G| = |H| [G:H]$$

$\therefore H$ 的阶一定是 G 的阶的因子. $\therefore 6$ 的因子有 1, 2, 3, 6.

$\therefore H$ 的阶可能是 1, 3, 2, 6

唯一性: 设 H, K 是 2 个不同的子群

$\therefore H \cap K$ 是 G 的子群 $\therefore H \cap K$ 也是 H 的子群

$\therefore$ 由拉格朗日定理 $|H \cap K| \mid 3 \therefore |H \cap K| = 1$ 或 3 若 $|H \cap K| = 3$ 则 $H = K$ 与 H, K 不同矛盾

$\therefore |H \cap K| = 1 \quad |H| = |K| = 3, H \neq K, H \cap K = \{e\}$

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{3 \times 3}{1} = 9$$

但 $HK \leq |G|$, 而 $|G| = 6 \therefore |HK| \leq 6$ 矛盾. $\therefore$ 不可能有 2 个不同的 3 阶子群

G 的非单位元的阶只可能为 2, 3, 6, 倘若 G 不含有 3 阶和 6 阶的元素, 即 G 除单位元外其它元素阶都为 2, 则 G 为交换群. 任取 G 的 2 个非单位的元素 a, b .

则 $H = \{e, a, b, ab\}$. 构成 G 的一个 4 阶子群. 由拉格朗日定理 4 | 6 矛盾.

$\therefore G$ 必含有 3 阶或 6 阶的元素. $\therefore$ 必存在 3 阶的元素

$\therefore$ 一定都有 3 阶子群.

4. $\because K/H \leq G/H \therefore$ 任意 $gH \in G/H, kH \in K/H$. 有 $(gH)(kH)(gH)^{-1} \in K/H$.

$$(gH)^{-1} = g^{-1}H$$

$$\therefore (gH)(kH)(g^{-1}H) = (gkg^{-1})H \in K/H$$

$$\therefore \text{存在 } k \in K. (gkg^{-1})H = kH$$

$$2 \uparrow \text{陪集相等.} \therefore gkg^{-1} \in kH.$$

$$\therefore \exists h \in H. \text{ 使 } gkg^{-1} = k.h$$

$$\because H \leq K \leq G. \therefore k, h \in K.$$

$$\therefore gkg^{-1} = k, h \in K.$$

$\therefore K$ 是 G 的正规子群.



5. $G = (\mathbb{Z}_{12}, \oplus_{12})$. $H = (\mathbb{Z}_8, \oplus_8)$.

$G = \langle 1 \rangle$. 有限循环群. $\therefore$ 同态 φ 由 $\varphi(1)$ 完全决定.

$\varphi(1) = h \in H$. $\because$ 在 G 中. $12 \equiv 0$. $\therefore \varphi(12) = 12h \equiv 0 \pmod{8}$.

$12h \equiv 0 \pmod{8} \Rightarrow 4h \equiv 0 \pmod{8}$. $\therefore 8 \mid 4h$. $2 \mid h$.

$\therefore h = \{0, 2, 4, 6\} \pmod{8}$.

$\varphi(1)$	$\varphi(n)$	$\ker \varphi$
0	0	$\mathbb{Z}_{12}$.
2	$2n \pmod{8}$.	$\{0, 4, 8\}$
4	$4n \pmod{8}$	$\{0, 2, 4, 6, 8, 10\}$
6	$6n \pmod{8}$.	$\{0, 4, 8\}$.

